

Trading Fire: the Arms Trade Network and Civil War

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Abstract

The last fifty years have seen two big changes in world politics. First, the most important violent conflicts now largely play out within states rather than between great powers. Second, the decrease in transportation cost has pulled even the smallest and remote countries in the the global exchange of goods and services. In this paper we study how these two fundamental elements of modern world politics interact by analyzing the effects of the trade in small arms on the severity of civil war measured in terms of battle deaths. Using an instrumental variables approach we provide credible evidence that the trade in small arms increases the deadliness for combatants in civil war. Our results also show that sanctions and arms embargoes decrease the loss of combatant life. In addition, our estimation strategy implies an effect of markets and the arms trade network on the transmission of violence to civil war locations. In essence the results show that the arms trade produces a *law of conservation of violence*. As one civil war ends, the resulting changes in the international market leads other war torn countries' imports to increase, which in turn increases the number of casualties in ongoing civil wars.

Introduction

Political conflict within states is the leading source of politically motivated violence and death in the world today, and small arms and light weapons (SALW) are the primary cause of casualties in these conflicts (Klare 1999, Lacina and Gleditsch 2005).¹ The sale of SALW to conflict areas is big business for arms suppliers. SALW imports to civil war locations consisted of USD\$8 billion of US\$31.7 billion in total legal global SALW sales between 1992-2013, which is approximately 25% of all legal small arms trade.² Clearly, civil war generates a great deal of demand for SALW.

In spite of the fact that SALW are heavily traded in civil war locations and nearly all battle-related deaths in modern civil wars are caused by them,³ the effect of the arms trade on war-fighting remains an open question. While there is broad speculation that arms sales may increase the intensity of civil wars, the matter has not been settled because of data limitations and estimation challenges related to problems of causal inference. The challenge with identifying causality stems from the fact that ongoing civil war is a primary source of demand in the global SALW trade. While arms may affect warfighting, purchasing decisions depend on demand generated by expected and realized conflict. Such endogeneity bias limits inferences based on estimated associations alone.

In this paper, we use an instrumental variables modeling strategy to analyze the effect of SALW imports on the scale of violence in civil war and also to examine the diffusion of violence through arms markets and networks. We use data on the global legal SALW trade and battle deaths in civil wars to identify the effect of SALW imports on the intensity of

¹SALW include conventional hand-held small caliber weapons such as pistols and assault rifles as well as man-portable or vehicle-mounted weapons operated by a single person or small teams such as heavy machine guns, anti-aircraft and tank guns, rockets, and grenade launchers (Marsh 2010).

²These import values account for only legal arms imports and say nothing about black market sales or government-to-government arms transfers.

³By Klares (1999, p.20) count, 80-90 percent of all casualties in recent civil wars have been produced by such weapons.

violence in civil wars. We show that arms trade networks influence the spread and rise of violence in civil wars. The results show that arms imports to ongoing civil wars lead to more battle deaths in those locations. Additionally, there is evidence that arms trade networks may channel violence from recently terminated civil war locations to war zones elsewhere and the effect of these market shocks from recently terminated civil wars depend on the diversity of civil war countries' trade connections in the arms trade network.

In our analysis, we use civil war terminations around the world to instrument for imports of SALW to locations with ongoing civil wars. Terminations of external civil wars are plausible instruments for arms imports, because the cessation of violence in a civil war elsewhere reduces aggregate demand for arms in global markets and causes arms to be more affordable in countries where war is ongoing.

Our findings confirm that arms imports to civil war locations in fact lead to increases in the intensity of violence. We find that a 1% increase in arms imports to countries that are fighting civil wars results in an increase of approximately 2 additional battle deaths in a given civil war. This is a sizable effect considering that the average variation in arms imports to war countries results in a shift in approximately 38 deaths, which is sufficiently large number of yearly battle-related deaths for a conflict to meet one commonly used definition of a civil war.⁴ In addition, terminating a civil war reduces global violence by about 3500 but the flood of arms to other conflicts increase violence in ongoing civil wars by 386, conserving % 12 of the battle deaths.

Our analysis adds to a growing body of research that examines the causes of the intensity of violence in civil war. Recent research identifies both economic and political factors that are associated with battle deaths in civil wars. Civil conflicts are more severe where there is more economic grievance (Lu and Thies, 2011) and lower incomes (Gleditsch, Hegre and

⁴For a conflict to be admitted as a civil war in Uppsala Conflict Data Program (UCDP) datasets, there must be at least 25 battle related deaths in a calendar year. We use UDCP data for all conflict-related measures throughout our analysis.

Strand, 2009; Lacina, 2006). When looking just at ethnic conflicts, recent work finds that globalization increases the severity of civil wars (Olzak, 2011). Political factors that affect the intensity of violence include regime type and the size of the governing coalition. Lacina (2006) finds that civil wars involving democracies result in fewer fatalities than those involving non-democracies, and Heger and Salehyan (2007) shows that smaller governing coalitions lead to a greater number of battle deaths. In non-ethnic conflicts, corruption increases battle deaths (Olzak, 2011). In addition to economic and political factors, scholars have also found that the deadliness of a conflict depends on geography and natural resource wealth (Lujala, 2009), ethnicity (Lacina, 2006; Eck, 2009), and the duration of the conflict (Lacina, Gleditsch and Russett, 2006; Melander, Oberg and Hall, 2006).

One contribution of our findings to existing work on civil warfighting is the potential for policy prescriptions for reducing violence. Most obviously, sanctions and arms to civil war countries may straightforwardly reduce violence in those locations. Additionally, the results suggest that market integration for civil war countries may mitigate the susceptibility of a country to market shocks from civil war outcomes elsewhere.

Our analysis also contributes to an existing research on civil war contagion and network diffusion. Studies on the spread of intrastate conflict focus primarily on spillover of the onset or ongoing violence within states (Lane, 2016) or in immediate civil war neighborhoods (Beardsley, 2011). Scholars have identified several mechanisms that account for diffusion across borders into contiguous states. These factors include state capacity and refugee flows (Braithwaite, 2010; Salehyan and Gleditsch, 2006*a*), ethnic and religious ties (Metternich, Minhas and Ward, 2015; Buhaug and Gleditsch, 2008; Fox, 2004), and ties to separatist movements (Buhaug and Gleditsch, 2008).

Our study adds arms networks to the list of mechanisms that facilitate the diffusion of civil conflict. However, our analysis differs in two critical ways. First, research on civil war contagion focuses on the spread of violence due to the onset of civil war or ongoing violence

within civil war. We show that civil war termination also contributes to the spread of violence. Second, in standard accounts, contagion is mostly relevant in local neighborhoods, but we demonstrate that contagion through trade networks may spread beyond state borders and immediate interstate neighborhoods. Because some countries are geographically distant but closely connected via a network pathway, we show that political and economic shocks can lead to the spread of violence beyond immediate neighborhoods.

In focusing on diffusion of violence through trade networks, our analysis ties into a large body of theoretical work that examines how network factors facilitate the transmission of shocks. These studies evaluate features of shocks such as frequency and location of impact within the network (Gai and Kapadia, 2000) as well consequences for diffusion depending on network factors such as interconnectedness, integration, and component size (Elliott, Golub and Jackson, 2014; Cabrales, Gottardi and Vega-Redondo, 2014; Goyal and Vigier, 2014; Allen and Gale, 2000). We apply these insights to assess empirically the extent to which the factors that we identify in the structure of the small arms network facilitate the transmission of shocks through the trade of small arms. In particular, we assess the effect of measures of degree connectedness for civil war arms importers and the network closeness to war termination shocks.

Our analysis gives insight into how markets may channel and conserve violence. We show that civil wars attract SALW imports, and the increase in weapons intensifies violence in those wars. Intuitively, we might then conclude that terminating civil wars straightforwardly reduces global violence. When a civil war ends, arms imports to the war torn areas decrease significantly, and battle deaths do in fact stop or decrease in that location. However, our analysis shows that violence from recently terminated civil wars is at least partially conserved through the global arms trade. The reduction of demand for weapons in countries where war has just terminated results in more weapons available in the global marketplace at lower prices. Locations with ongoing civil war, where demand for weapons remains relatively high,

import more weapons at the lower prices and, as a result, violence increases in these wars.

In addition to suggesting a notion of partial global conservation of violence, our findings also identify a novel form of contagion of conflict. Our analysis focuses on the cessation of violence and shows that the termination of civil war in a country may lead to the spread of violence to many conflict-prone countries around the world. This unique pathway of contagion raises new questions about the interconnectedness of global politics and economics, such as the possibility that policies designed to influence civil wars or to manipulate global arms markets may have unanticipated effects on the use of violence elsewhere. Our results imply that even regional violent conflicts and the policies designed to address these problems are not isolated in their effects from the rest of the world. Under certain conditions, interdependencies in global markets may channel and spread political trends and violence in unexpected ways.

Arms Trade Networks and the Severity of Civil War

The intensity of civil wars vary a great deal both across different wars and across time within wars. Figure 1 shows the variation in yearly battle deaths in 51 civil wars that lasted 5 years or more between 1992 and 2013. It is noteworthy, that there does not appear to be any definite pattern to war fighting intensity. While battle deaths hold steady in several wars, the intensity of violence builds over time in other wars. Some conflicts are marked by an intense outbreak of violence that gradually levels off, and other wars reflect seemingly random fluctuations of battle deaths throughout the duration of the conflict.

We argue that the arms trade affects battle deaths in civil wars in two ways. The first argument is straightforward, and confirms prevailing wisdom regarding the influence of weapons in conflicts. We show that SALW imports lead to increases in battle deaths in ongoing civil wars. Fighters need weapons to produce violence. As weapons become more



Figure 1: Yearly Battle Deaths by ConflictID

available, violence in war increases.

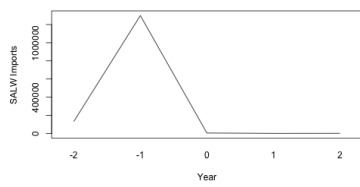
The second argument is more nuanced and speaks to existing studies on civil war contagion and network diffusion. Our claim is that shocks to the arms trade can lead to the spread of violence. Existing studies of civil war contagion focus primarily on the diffusion of the outbreak of violence to immediate civil war neighborhoods. We argue that civil war termination also plays a role in the spread of violence, and may do so in locations that are not necessarily in close geographic proximity to the recently terminated civil war country.

The argument focuses on market dynamics as mechanisms for signaling price shifts to arms importers, and network relationships as pathways for channeling arms to conflict zones

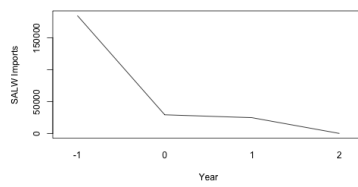
where violence is ongoing and can intensify with more weapons. The cessation of violence at the end of a civil war results in a drop in demand for weapons in the locations where war was recently being fought. When wars terminate, these countries no longer import as many weapons, and supplies of weapons previously used in the civil war become available for trade in the arms network. Figure 2 shows the logged value of SALW imports for a variety of countries that recently experienced a civil war. The graphs are centered on the year that the civil war ended. For all cases imports to locations with civil war terminations drop the year of a war termination. For nearly all the cases of terminated wars, SALW imports remain dramatically lower following war termination. For two countries, Guinea and Serbia, imports rose for one year before dropping permanently. In only one case, Ethiopia, do we see arms imports increasing starting one year after the war termination.

The spike of the weapons supplied to the market after a civil war terminates impacts aggregate demand and the prices of weapons. This, consequently, influences purchasing behavior in other locations where demand remains high because of political crisis and conflict. Countries with ongoing civil war should import more arms when countries elsewhere terminate civil wars. The impact of the shock of a civil war termination depends on geographic and network proximity. The effect of price shocks dissipates over geographic distance. We expect, then, that the closer a country with ongoing civil war is to a country that has terminated a civil war, the more it will import from that country.

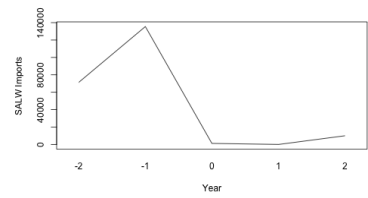
Geography, however, is not the only limiting factor. Countries that are geographically proximate may not trade directly with one another. In fact, the trading pathway between these two countries may have multiple other countries in between where arms shipments get routed before reaching their final destination at one of the two countries. Additionally, two countries that are geographically distant may have few, if any, countries that lie between them on the trading pathway. This geographically distant pair of countries is closer on the network pathway than the first pair of geographically proximate countries.



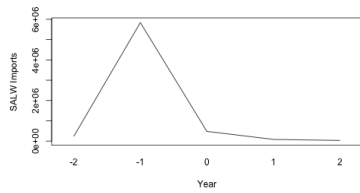
(a) Cambodia



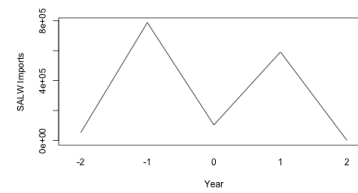
(b) Djibouti



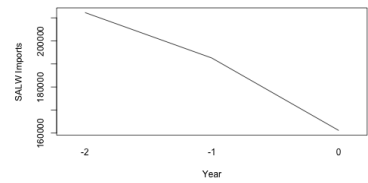
(c) Congo



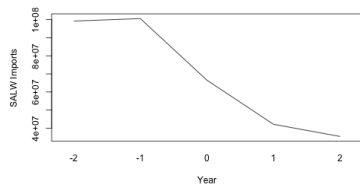
(d) Sri Lanka



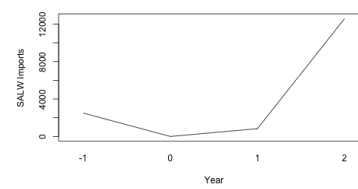
(e) Guinea



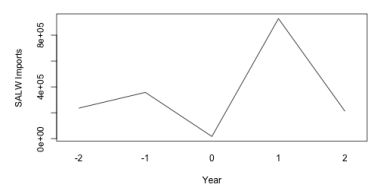
(f) Tajikistan



(g) United Kingdom



(h) Ethiopia



(i) Serbia

Figure 2: Time series of arms imports centered at war termination

To understand how network distance may relate to the transmission of pricing shocks due to terminated civil war, we lean on the intuition of theories of networks that show that interconnectedness and integration in networks facilitates the transmission of shocks throughout a network (Elliott, Golub and Jackson, 2014; Cabrales, Gottardi and Vega-Redondo, 2014; Goyal and Vigier, 2014; Allen and Gale, 2000). We argue that the impact of the civil war termination shocks depends on network connectedness. The closer an importer country is to another country with a terminating civil war on the arms trade pathway that connects the two countries, the greater the increase in arms imports to the importer country. Similarly, greater network distance leads to more arms imports, as we expect that market price dynamics will be muted as imports must make their way through multiple stops first before arriving at their end destination.

In many trade networks, especially the global trade of goods and services, there is little variation in network distance between dyads because such networks are densely connected compared to the SALW trade network. To see this, we generate network measures of the legal small arms trade using conventional classifications of SALW and inflation-adjusted values of arms imports and exports of small arms. These data are taken from the UN Comtrade data agglomerated by NISAT (Marsh, 2013). Reliable data on legal trade in small arms and light weapons are available from 1992-2013.

First, relative to other networks, the SALW trade network is not strongly connected. There are some pairs of countries who are not linked through any pathway of importers or exporters. Compared to the benchmark of the global trade network for goods and services, the small arms trade network is also significantly less dense. For example, if we measure the density of a network by the fraction of actual connections divided by the total number of possible connections, the density of the global trade of goods and services in 1992 is .5. By contrast, the fraction of total possible connections in the small arms trade is only .047. Thus, even though the arms trade consists of many active buyers and sellers, the weapons

themselves flow through relatively few established trade routes. Countries along these routes are influential players in the arms trade. Identifying the factors that predict the emergence of influential actors along these routes will improve our understanding of how third parties can influence the path of weapons to and from civil wars.

For example, we find that there are several countries that are highly connected and trade arms with many different countries. To determine whether exclusive or dependent trade relationships exist, we might wonder if these highly connected traders are also linked to one another. As a point of comparison, in the global trade network for goods and services, trade is so integrated that all countries, including highly connected countries, are connected with one another. This configuration means that there are relatively few bottlenecks or asymmetric dependencies in the global trade network for goods and services. By contrast, in the small arms trade network, highly connected traders tend not to be connected with one another, implying that some highly connected actors occupy exclusive sourcing positions with respect to clients. To determine whether countries with many trading partners tend also to trade with highly connected countries, we can compute the degree assortativity of the small arms trade. In 1992 the degree assortativity in the arms network is $-.48$ meaning that highly connected traders tend not to be connected to each other. This implies that the small arms network consists of a set of important players trading with relatively disconnected partners on the periphery.⁵

Given that SALW trade networks have relatively few trade pathways and variation in the distance of those pathways as compared to denser and more interconnected trade networks, we expect that network distance is likely to play a role in determining whether market shocks from civil war terminations are felt in countries with ongoing civil war. Thus, we expect that both geographic and network distance from terminated civil wars affects the amount of

⁵Further analysis of the network over time would show whether this is the product of the data's proximity to the end of the Cold War or whether this is a trait that persists.

arms that countries with ongoing civil wars import. The closer the geographic and network distance, the more arms imported. This implies that countries that are geographically distant from terminated civil wars may nevertheless still import more arms because it is closely connected with a direct or close network arms trade relationship.

When the effect of distances to civil war terminations on arms imports is combined with the first argument that increases in arms imports leads to more battle deaths in civil war countries, the overall evidence is suggestive of a larger narrative about the transmission of violence through arms markets and networks. Terminated civil wars transmit violence to conflict zones elsewhere through market actors who are incentivized to arbitrage on shifts in arms prices, which activity redistributes arms to areas with ongoing civil war. Thus, even though terminated civil wars reduce battle deaths in that theater, violence is at least partially conserved as it spikes in other theaters.

Empirical Strategy

Causal identification is a main challenge to analyzing the effect of the arms trade on the scale of violence in civil wars. Arms imports may affect warfighting, but wars generate demand for importing weapons. We use an instrumental variables approach to estimate the effect of imports on the intensity of civil war and to address the broader question of the role of arms networks in the diffusion of civil war.

We consider civil war terminations as a source of exogenous variation that affect decisions to import small arms into locations elsewhere with ongoing conflict. Ideally, we would look for a process in which arms market fundamentals change in ways unrelated to the demand associated with existing conflicts. Such an instrument would allow for credible estimation of the effects of arms imports on battle related casualties. We argue that the termination of other civil wars is a source of exogenous variation that satisfies the exclusion restriction.

As discussed above, arms imports decrease in locations where war has terminated. Take for example the the end of the civil war in Sri Lanka. In 2008, the year the government withdrew from the ceasefire, small arms and light weapons imports were US\$5.9 million. In 2009, the year the LTTE conceded defeat, Sri Lanka's small arms imports dropped to approximately \$476,000. In the termination year the imports were less than one-tenth the level the year before, thus substantially changing the market for arms in South Asia.

This example illustrates the path by which we identify the effect of arms sales on the severity of civil war as measured by battle deaths. At the end of a civil conflict weapons become less valuable and this results in a decrease in a country's imports and overall demand. A reasonable expectation is that the decrease in demand lowers the price for arms—changing the marginal price of purchasing weapons in other on going conflicts. If the other conflicts were previously operating in a market with a given price for small arms, this reduction in demand elsewhere will have ripple effects, changing prices and resulting in more purchases in other wars.

Though we may expect that the changes in the demand for arms from terminated civil wars affect other proximate civil wars (Bourne, 2010), there is strong evidence of weapons from various civil wars – like the Balkan War – showing up in conflicts around the world. Thus local changes in demand can have far reaching effects, though we might reasonably conclude that they are diminishing with both geographic and network distance. Thus, we construct two instrumental variables that measure the weighted distances of a country with ongoing civil war to countries that have terminated civil war.

The first variable, **GEOGRAPHIC CLOSENESS WEIGHTED TERMINATION**, is the sum of one over the geographic distance of terminations in a given year for each country involved in an ongoing civil war. We construct this variable by using distance data from the C-shape database (Weidmann, Kuse, and Gleditsch 2005), which measures the distance between capital cities of countries through 2011. We measured the capital to capital distance from

the location of each country in which there was a civil war termination to every primary civil war location. We use these distances to generate a “weighted” shock to demand of arms ongoing civil conflict, giving more weight to nearby terminations as regional markets are known to be important in the small arms trade (Bourne, 2012). We do this for each country with an ongoing civil war by calculating one over the geographic distance to a country with a terminated civil war in a given year. We do this for all a civil war terminations in a given year and then sum the distance weights between the country with an ongoing civil war and each civil war termination in a given year.⁶

For the second instrumental variable, NETWORK CLOSENESS WEIGHTED TERMINATION, we repeat the process above but use directed geodesic distance between dyads in the SALW trade network rather than geographic distance. For a country with ongoing civil war, we calculate the shortest directed geodesic distance between that country and each country with a civil war termination in a given year. We then take the sum of one over this value for all of the civil war terminations in a given year.

We estimate a two-stage GMM model. Geographic and network distance, as measured by GEOGRAPHIC CLOSENESS WEIGHTED TERMINATION and NETWORK CLOSENESS WEIGHTED TERMINATION, is used to instrument for logged SALW imports in the first stage. ConflictID fixed effects are included to capture time-invariant country characteristics. The equation we estimate for the first stage is:

$$\text{LOG SMALL ARMS IMPORTS}_{it} = \mu + \theta_1 G_{it} + \theta_2 N_{it} + \delta X_{it} + \eta_{it}.$$

Small arms imports is measured by taking the log of imports for small arms and light weapons for conflict county i in year t , G is the geographic closeness weighted termination score or the distance weighted sum of civil war terminations for each ongoing war, N is the the network

⁶This distance measure is the most commonly used in geography. (cite needed)

closeness termination score or the directed geodesic weighted sum of civil war terminations for each ongoing war, X is a matrix of “control variables” that include conflict fixed effects, and η_{it} is country year specific error term.

The second stage model estimates the causal effect of imports on severity of conflict. Here we estimate, using the instrumental variables for small arms, the following equation:

$$\text{BATTLE DEATHS}_{it} = \alpha + \beta \text{ LOG SALW IMPORTS}_{it} + \gamma X_{it} + \epsilon_{it}.$$

Battle deaths are total number of battle deaths in conflictid i in year t . X_{it} is a series of controls including conflict fixed effects, and ϵ_{it} is a conflict year specific error term. To satisfy the exclusion restriction for our analysis, we further need

$$\text{cov}(G, N, \epsilon_{it} | X_{it}) = 0.$$

That is, conditional on the controls the covariance between the instrument and the error term in the second stage needs to be zero.

We consider two possible violations of the exclusion restriction. First, terminations of civil war may make foreign fighters available to participate in ongoing conflicts elsewhere. Second, termination of internationalized civil wars may lead intervening countries to redirect their efforts to other ongoing civil wars. In robustness checks, we control for both foreign fighters and internationalized civil wars. Additionally, we run a sensitivity analysis that tests the contagion hypothesis.

Data

We construct a dataset of civil wars, battle deaths, and UN Comtrade customs records of legal interstate SALW trade from 1992-2013. The data is structured as a civil war panel that

describes factors related to arms flows and violence for each year in the duration of a civil war. An observation is a year within a unique civil war. We include all internal civil wars from 1992-2013 taken from the Uppsala Conflict Data Program (UCDP) Armed Conflict Data v4 (2014).⁷ If a war began prior to 1992 but was ongoing between 1992-2013, then we include the relevant years in the dataset. The years between 1992-2013 of civil wars that had not terminated by the end of 2013 are also included in the data.

We include internationalized and non-internationalized intrastate wars because some conflicts change from non-internationalized to internationalized or vice versa. We do not want to exclude war years based on this distinction and then inadvertently miscode transitions across types of conflict as the beginning or end of episodes of conflict. We include a variable for war years in which conflicts are internationalized so that we can control for external intervention in the analysis. Our dataset includes a total of 114 ongoing civil wars between 1992-2013 and a total of 906 total civil war years across these ongoing civil wars.

We are mainly interested in estimating the effect of small arms imports to all battle locations in a civil war on the severity of violence in these wars. Severity of civil wars is measured by the number of battle deaths recorded annually in a civil war according to best estimates from the UCDP Battle-Related Deaths Dataset v.5 (Lacina and Gleditsch, 2005).

Our measure of arms imports is an inflation-adjusted value of small arms legally imported into a battle location of an ongoing civil war. These data are taken from the UN Comtrade data agglomerated by NISAT (Marsh, 2013). We include imports of small arms and light weapons only and exclude larger conventional weaponry and heavy artillery. Reliable data of small arms and light weapons are available from 1992-2013. The value of arms is reported in US dollars and adjusted for inflation in 2013 dollar values.

A complication with organizing the data is the fact that some internal conflicts spill across international borders resulting in battles in multiple countries, all of which potentially

⁷ *Uppsala Conflict Data Program. UCDP Conflict Encyclopedia* (Date of retrieval: 08/14).

import small arms, for one civil conflict. To create a measure of the total value of legal arms imported to a civil war in a given year, we first identified all of the countries identified by the UCDP/Conflict data as serving as battle locations for a intrastate war in a given year. We then summed the reported yearly value of small arms imports to each of these country locations to give the annual value of arms imports to a civil war.

Our arms trade data does not reflect all small arms sales to a civil war. Because we only have access to information about legal imports that were recorded by UN customs statistics, our data does not include illicit arms or legal arms that were not reported by customs. There are some additional limitations to the observed trade data. Customs data relies on self-reporting by exporters and importers, so the reliability of the value of the arms in the data depends on the accuracy of these reports. Additionally, these reports only document the transfer of shipments across borders. Consequently, it is not known what percentage of reported imports flow to the government, rebel groups, or get routed illegally to end destinations outside the country. These are all known limitations of the data. Nevertheless, these data are currently the best description of the arms trade available, and there is reason to be confident that they provide a fairly accurate approximation of the arms trade in spite of the known limitations. According to the 2012 Small Arms Survey ⁸, the known small arms trade is worth approximately US\$8.5 annually, nearly all of which is accounted for by legal Comtrade data. The best estimates project that the unknown trade may add as much as US\$1.5 billion to that amount. Thus, it is believed that the reported legal small arms imports in our data account for a significant percentage of the global arms trade.

In addition to this main variable, we also include data on several additional variables which may be associated with violence in civil wars. In particular, we include measures of development aid, income, and regime type for the primary government in the conflict. Income data is a standard measure of per capita GDP as reported by the recent release

⁸*Small Arms Survey: Moving Targets* (2012).

of the Penn Tables gdp data (v8.1 2015), which runs through 2011. To measure regime type, we include polity scores of the primary government from the Polity IV data project (v2014) (Marshall, Jaggers and Gurr, 2002). Polity Scores measure regime authority on a 21-point scale ranging from -10 (hereditary monarchy) to +10 (consolidated democracy) and are available through 2013. We include the amount of foreign aid received by a country from the AidData project (v3.0, data runs through 2011). The aid values are in constant 2009 US\$. We also include bilateral total global trade data (Barbieri v3.0 2012 release, data runs through 2009), which includes imports and exports for each country. The global trade data is limited in that it only runs through 2009. Measures for countries capabilities are taken from Correlates of War Capabilities Index (CINC scores v4.0) and run through 2007. Finally, we include an indicator variable for the presence of foreign fighters in a civil war taken from Malet (2011).

Results

The evidence that civil war terminations affect arms imports extends beyond just the figures above and the two examples.

Table 1: Effect of Weighted Geographic and Network Closeness of Foreign Civil War Terminations on Arms Imports, 2-stage GMM Imports in Ongoing Civil Wars

	SALW Imports First Stage
Geographic Closeness	53.97***
Weighted Termination	(3.82)
Network Closeness	0.460***
Weighted Termination	(4.63)
N	670
Kleibergen-Paap Wald F-Stat	17.45

t statistics in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Conflict FE, Heteroskedastic and Autocorrelation Consistent SE, Bartlett Kernel Bandwidth 3

Table 1 presents the first stage of the two-stage GMM used in our analysis. The model is the simplest specification, a regression of the geographic closeness weighted termination score and network closeness weighted termination score for each civil war country on their SALW imports with conflict fixed effects. The effect is substantial and statistically significant for both distance scores. The Kleinbergen-Papp Walk F-Statistic is also large enough at 17.45 to reject the null hypothesis of weak instrumentation.

In our data the average GEOGRAPHIC CLOSENESS WEIGHTED TERMINATION is .001 and has a standard deviation of .003. The maximum value is .06. To get a sense of the substantive effect of geographic closeness instrument from the first stage analysis, a one standard deviation change in a country's termination score is 16.19% change (.002x100x61.02) in small arms imports. The average NETWORK CLOSENESS WEIGHTED TERMINATION is 1.42 and standard deviation is 1.15. A one standard deviation change in a country's network closeness instrument results in a 52.9% change in small arms imports.

For the full two-stage GMM model we have a second stage estimation to obtain the causal effect of imports on severity of conflict. Table 2 presents the results for three models. In column (1) we present the simple OLS regression, with no instrument, of BATTLE DEATHS on SALW and country fixed effects. Here we see a rather small and insignificant effect of arm imports on the number of battle deaths. Specifically the estimate shows that a one-percent increase in small arms and light weapons imports accounts for about 0.3 battle deaths in a given year. But the result is not statistically significant.

In column (2), we run the same model specification, but with our instruments. The effect of small arms imports is significant and over seven times as large. Here a one percent increase in small arms imports leads to over two new battle deaths.

Column (3) controls for GDP per capita and the polity score of the civil war's primary location. The estimated effect of small arms imports remains significant and quantitatively stable at about two battle deaths per percentage point increase.

Table 2: Effect of Imports on Battle Deaths in Ongoing Civil Wars, 2-stage GMM

	(1) OLS Model	(2) Second Stage	(3) Second Stage
SALW (logged)	31.19 (20.81)	222.6** (78.03)	200.8* (96.11)
GDP per cap (logged)			-898.7** (343.0)
Polity			-33.60 (20.37)
N	670	670	534
K-P Wald F-Stat		17.45	13.00

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Conflict FE, Heteroskedastic and Autocorrelation Consistent SE, Bartlett Kernel Bandwidth 3

Robustness

Table 2 provides a series of estimates and diagnostics that link small arms imports and the severity of civil war as measured by battle deaths. The validity of the instrumental variable analysis, however, depends on the assumption that our termination score instrument has no direct effect on the severity of civil wars in other countries—that is that the instrument is exogenous.

Though the assumption has face validity we can further substantiate our results by looking at additional models that include variables that could be plausibly associated with the termination scores and battle deaths.

Table 3 presents six additional models with a variety of new control variables. Columns (1) and (2) are models where we include other measures of economic connectedness. Model one adds a control for foreign aid receipts as taken from AidData 3.0. Foreign aid is negatively associated with battle deaths, but we cannot be sure that it is not severity of civil war that decreases aid or the other way around. But what we do see is that if one were worried

that the termination of other civil wars changed the international distribution of foreign aid and hence the severity of continuing civil wars, the effect of small arms and light weapons remains in the expected direction.

Model two in column (2) adds total trade to the analysis. The result for small arms remains positive and significant at the 0.001 percent level, though the number of observations drop.

Table 3: Effect of Imports on Battle Deaths, fe robustness

	(1)	(2)	(3)	(4)	(5)
SALW (logged)	191.9 (106.5)	231.9*** (67.99)	182.3* (89.92)	210.0* (99.13)	191.7** (68.96)
GDP per cap (logged)	-805.2* (339.7)	159.7 (504.5)	-807.7* (315.7)	-886.8** (337.0)	215.3 (510.6)
Polity	-33.57 (20.64)	-23.29 (21.56)	-35.27 (20.08)	-33.56 (21.16)	-29.23 (21.53)
Foreign Aid (logged)	-84.33* (41.50)	-100.7* (48.75)			-109.9* (50.78)
Total Trade (logged)		-415.3* (166.3)			-369.5* (167.7)
Internationalized			-493.9 (349.3)		-542.7 (412.0)
Foreign Fighters				-217.5 (268.3)	-383.0 (294.4)
N	514	465	534	513	465
K-P Wald F-Stat	11.69	14.18	13.42	11.71	14.31

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Conflict FE, Heteroskedastic and Autocorrelation Consistent SE, Bartlett Kernel Bandwith 3

Columns (3) and (4) report results from models that control for what might be alternative pathways for the termination of other civil wars to influence the number of battle deaths in

an ongoing civil war. The first variable we consider is whether or not the civil conflict is internationalized according to the UCDP code when internal armed conflict occurs between the government of a state and one or more internal opposition groups with intervention from other states (secondary parties) on one or both sides. Like in the case of aid, one might imagine that as other civil wars terminate resources of third parties may be freed to be redeployed to ongoing civil wars. As such wars that are internationalized may receive these additional resources that, in turn, may effect the number of battle deaths directly. As is seen in column (4), internationalized conflicts have no discernible effect on the number of battle deaths in a year and the estimate of the effect of small arms imports are stable and robust.

Second we consider the possibility that foreign fighters may affect battle deaths and as they become free from obligations and duties in a terminated civil conflict they may move and have a direct effect on the number of battle deaths in ongoing wars. Coding data found in (Mallet 2013) we generate an indicator variable for the presence of foreign fighters in the primary civil war location in a given year. Again there is not effect of foreign fighters distinguishable from zero. Small arms and light weapons have the same substantive and significant effect.

Column (6) includes all variables. SALW remains significant at nearly the five percent level, though its effect drops slightly compared to the other specifications with no to minimal controls. That said, it is still six times the size of the OLS estimate.

Table 4 presents the results from a sensitivity analysis that tests the contagion hypothesis. In the contagion argument, civil conflict spreads, through a number of mechanisms—such as refugees (Salehyan and Gleditsch, 2006*b*). Here we assume that direct effect of GEOGRAPHIC DISTANCE WEIGHTED TERMINATION and NETWORK DISTANCE WEIGHTED TERMINATION can be positive or negative, i.e., that inasmuch as war is contagious, peace is similarly contagious through the inverse of the same mechanisms. We start by assuming that the direct effect of GEOGRAPHIC DISTANCE WEIGHTED TERMINATION is between 100 and -

Table 4: Union of Confidence Intervals with IV Direct Effect between -100 and 100.

Variable	Lower bound	Upper bound
LOG SALW	28.198366	349.28

100 and NETWORK DISTANCE WEIGHTED TERMINATION is between -2 and 2. Then we estimate 1000 models on the grid, and take the union over their 95% confidence intervals. For each LOGGED SALW variable in model, the lower bound in Table 4 is the minimum of all the left-hand side confidence intervals across our 1000 models. The upper bound, in the third column, is the maximum of the right-hand side confidence intervals of all 1000 models.

The analysis shows that the effect of small arms imports is still distinguishable from zero. In fact we can say more. If we thought that the direct effect of the weighted terminations variables were positive, the union of 95% confidence intervals would be distinguishable from 0 until its coefficient is greater than 100 and 2 (respectively). We note that this is a hard test in the sense that it treats every estimate of the direct effect, even the most extreme guesses, as equally likely.⁹

Discussion

While the size and effect of the arms trade on civil war severity as measured by battle deaths is reasonable on its face, it is important to be able to demonstrate the substantive effect of arms imports. On average models two and three predict 2.1 battle deaths for every one percent increase in arms imports. In an average year of an on going civil war there is a 13 percent change in the level of imports and at 2.1 battle deaths per percentage point, changes in imports account for, on average, 27.3 battle deaths, enough to get a civil war entered into

⁹If we thought that extreme values were less likely than small values, then the union of these Bayesian credibility intervals would be even smaller.

the UCDP conflict dataset.

redo: At one standard deviation above the mean, the average change in the small arms trade is 35 percent. By our estimate the small arms trade would account for 101 battle deaths in such civil war years and would be sufficient for a conflict crossing the threshold to be entered into the Correlates of War dataset.

To judge the effects of the small arms trade we can also look at particular transactions. For example in 1992 South Korea exported \$23,052 worth of small arms to Serbia, about 8% of Serbian small arms imports that year, and by our estimates resulted in 23 battle deaths.

We could also think about the positive effects of sanctions in this light. In cases like Liberia, Eritrea, and the civil war in south Sudan, the international community interventions can decrease the loss of life. For example in May of 2000 UNSC resolution 1298 embargoed the shipment of arms and ammunition to Eritrea. Between 2000 and 2001 the arms imports of Eritrea fell by 99%, which would result in the 287 fewer casualties in that year.

We can also quantify another implication of our analysis – that a decrease in violence that results from the termination of one conflict is partially off-set by the transfer of arms from the terminating conflict to new conflicts. For instance, take the wars of independence of the Former Yugoslavia. Many arms flowed into this conflict from sources ranging from the former Soviet Union to Muslim states, such as Iran, in the middle east. Then in 1998 Croatia exported \$1,000,000 in arms to Colombia, representing a 10% increase and implying 29 additional battle deaths in that year. This single transaction accounted for 5 percent of casualties in the that year and was large enough to give Columbia a sufficient number of battle-deaths to qualify as an entry in the UCDP dataset.

As we see that the substantive effects of the small arms trade on the level of violence is important, it also illustrates how the international market for small arms and light weapons perpetuates violence. Our instrumental variable analysis shows that stopping wars in one location leads more violence in other locations. We can see this effect results in a conservation

of violence as a result of the arms trade. How might we get a sense of the coefficient on the conservation of violence? Consider a simple calculation based on data in our sample. If the average number of civil war terminations in a year is approximately 4 and the average number of battle deaths is 837, then the termination of civil wars, by whatever means, decreases battle deaths by 3348 on average. On the other hand, an average TERMINATION SCORE in a year is .001, multiplied by the coefficient in first stage of our analysis (100×61.02), itself multiplied by the coefficient on the Log of SALW imports from the second stage ($.01 \times 297$), can then be multiplied by the average number of ongoing wars (37). We thus estimate that in the average year the small arms trade resulting from civil war terminations leads to about 670 new battle deaths. That is, the effect of trade resulting from civil war terminations in an average year is good for 3/4 of a new civil war worth of battle deaths.

Conclusion

Most of the damage in modern civil wars is done by small arms, like pistols, machine guns, grenades, and mines. By Klare's (1999, p.20) count, 80-90 percent of all casualties in recent civil wars have been produced by such weapons. As pointed out by Boutwell and Klare (2000), small arms are the weapon of choice in civil conflicts because they are readily obtainable, cheap, deadly, and easy to transport. It is also the case that large stockpiles of small arms exist from previous wars, including the Cold War, and unlike major weapons systems, there exists an easily accessible market for their purchase outside the direct control of states. This is particularly important when the groups looking to arm might be classified as outlaws and pose threats to their own state or neighboring regimes.

Though there are many factors that contribute to the onset and severity of civil war, many of them are outside the control of anyone. On the other hand, practitioners argue both that moving arms into a conflict can deter one-sided casualties and that embargoes

are important tools for reducing the deadly nature of civil war. Here we show there are significant effects of the trade in small arms on the severity of civil war, measured in terms of battle deaths. From our analysis we can conclude that the effect is positive and that sanctions and arms embargoes decrease the loss of combatant life. Second, our estimation strategy implies an effect of markets and trade networks on severity of violence. In essence the results show that networks transmit market shocks to impact warfighting in conflict zones removed from the source of the shock. We see that mechanisms of civil war contagion that are not limited to immediate civil war neighborhoods and the outbreak of violence. Close network connections may facilitate the spread of violence beyond immediate neighborhoods, and civil war terminations, in addition to civil war onset, may lead to the diffusion of civil war. Indeed, the analysis here suggest the possibility that various different phenomena may trigger the spread of violence insofar as those phenomena trigger market shocks that transmit price fluctuations through network pathways. Finally, the arms trade produces a partial *law of conservation of violence*. As one war ends, the incentives to sell unused arms on the international market leads other war torn countries' imports to increase and this increases the casualties in other theaters of violence.

References

- Allen, Franklin and Douglas Gale. 2000. "Financial Contagion." *Journal of Political Economy* 208(1):1–33.
- Beardsley, Kyle. 2011. "Peacekeeping and the Contagion of Armed Conflict." *The Journal of Politics* 73(04):1051–1064.
- Bourne, Mike. 2010. "Arming conflict: The proliferation of small arms."
- Bourne, Mike. 2012. Small Arms and Light Weapons Spread Conflict. In *Small Arms, Crime and Conflict: Global Governance and the Threat of Arms Violence*, ed. Owen Greene and Nicholas Marsh. Routledge Press pp. 29–42.
- Boutwell, Jeffrey and Michael Klare. 2000. "The Scourge of Small Arms." *Scientific American* .
URL: <http://amacad.org/news/scourge.aspx>
- Braithwaite, Alex. 2010. "Resisting Infection: How State Capacity Conditions Conflict Contagion." *Journal of Peace Research* 47(3):311–319.
- Buhaug, Halvard and Kristian Skrede Gleditsch. 2008. "Contagion or Confusion? Why Conflicts Cluster in Space." *International Studies Quarterly* 52(2):215–233.
- Cabrales, Antonio, Piero Gottardi and Fernando Vega-Redondo. 2014. "Risk Sharing and Contagion in Networks." *Unpublished* .
- Elliott, Matthew, Benjamin Golub and Matthew O. Jackson. 2014. "Financial Networks and Contagion." *American Economic Review* 104(10):3115–3153.
- Fox, Jonathan. 2004. "The Rise of Religious Nationalism and Conflict: Ethnic Conflict and Revolutionary Wars, 1945-2001." *Journal of peace Research* 41(6):715–731.

- Gai, Prasanna and Sujit Kapadia. 2000. "Contagion in Financial Networks." *Proceedings of the Royal Society A* 466(2120):2401–2423.
- Goyal, Sanjeev and Adrien Vigier. 2014. "Attack, Defense, and Contagion in Networks." *Review of Economic Studies* 81(4):1518–1542.
- Klare, Michael. 1999. "The Kalashnikov age." *Bulletin of the Atomic Scientists* 55(1):18.
- Lacina, Bethany and Nils Petter Gleditsch. 2005. "Monitoring Trends in Global Combat: A New Dataset of Battle Deaths." *European Journal of Population/Revue européenne de Démographie* 21(2-3):145–166.
- Lane, Matthew. 2016. "The Intrastate Contagion of Ethnic Civil War." *The Journal of Politics* 78(2):396–410.
- Marsh, Nicholas. 2013. *Database of Authorised Transfers of Small Arms and Light Weapons*. NISAT Small Arms Trade Database. Oslo: Norwegian Initiative on Small Arms Transfers.
- Marshall, Monty G, Keith Jaggers and Ted Robert Gurr. 2002. *Polity IV project*. Center for International Development and Conflict Management at the University of Maryland College Park.
- Metternich, Nils W., Shahryar Minhas and Michael D. Ward. 2015. "Firewall? or Wall on Fire? A Unified Framework of Conflict Contagion and the Role of Ethnic Exclusion." *Journal of Conflict Resolution*, forthcoming.
- Salehyan, Idean and Kristian Skrede Gleditsch. 2006a. "Refugees and the Spread of Civil War." *International Organization* 60(02):335–366.
- Salehyan, Idean and Kristian Skrede Gleditsch. 2006b. "Refugees and the spread of civil war." *International Organization* 60(02):335–366.

Small Arms Survey: Moving Targets. 2012. Cambridge University Press.

Uppsala Conflict Data Program. UCDP Conflict Encyclopedia. Date of retrieval: 08/14.

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URL: *www.ucdp.uu.se/database*